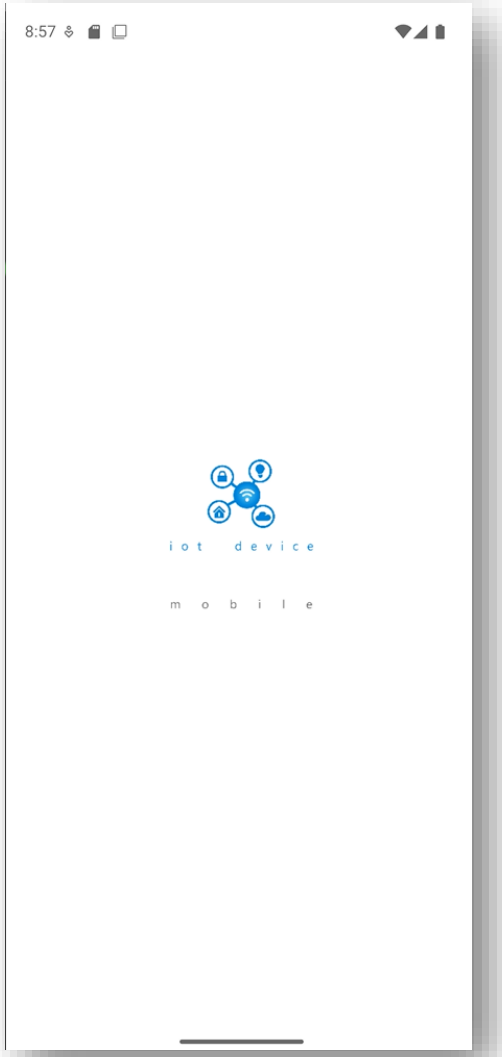
IoT Solutions

Development – Simulators

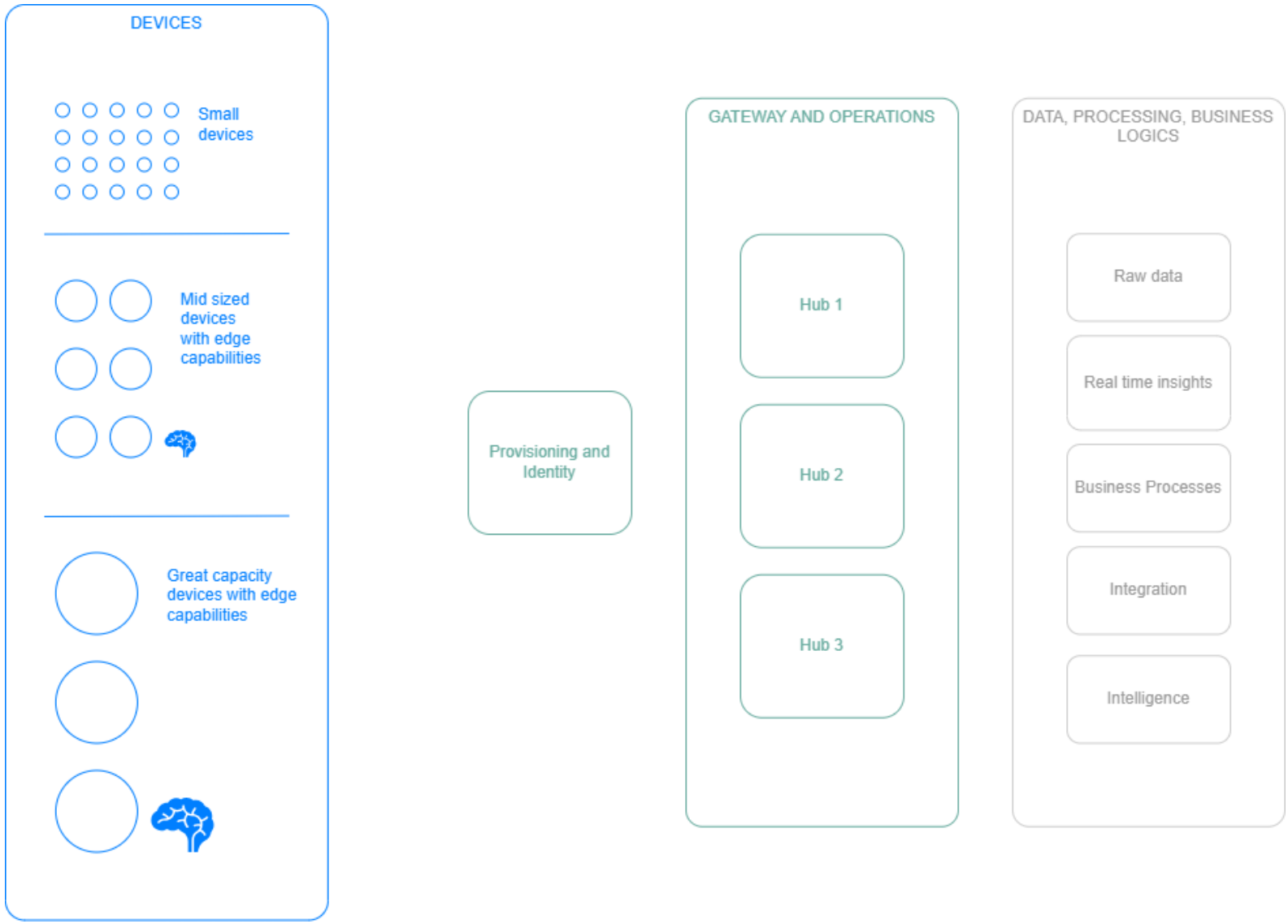
Jon Mikel Inza



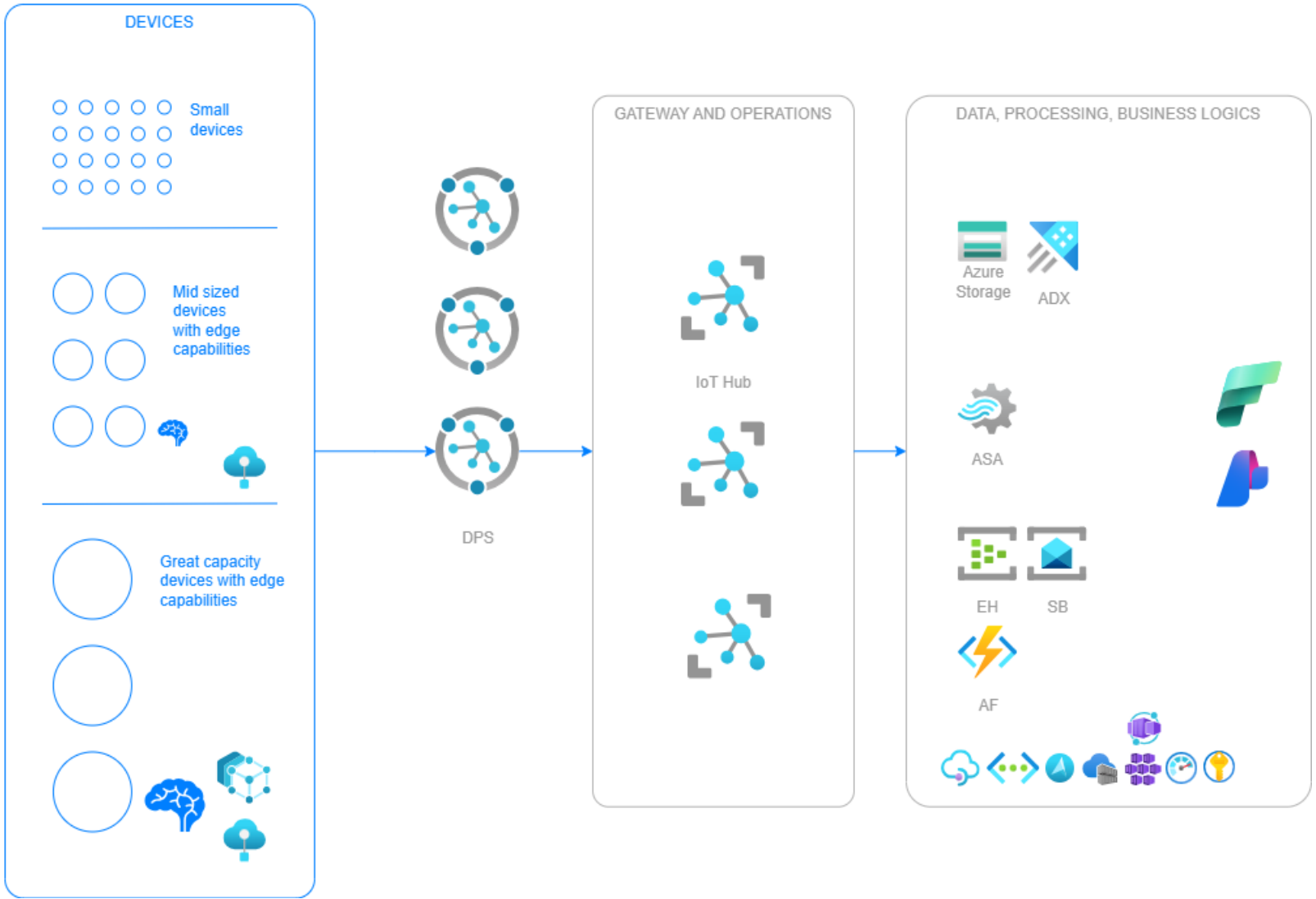
Agenda

- IoT flows and solutions (reminder)
- Azure IoT main services
- Why a device simulator tool ?
- Mobile IoT device simulator
- Q/A

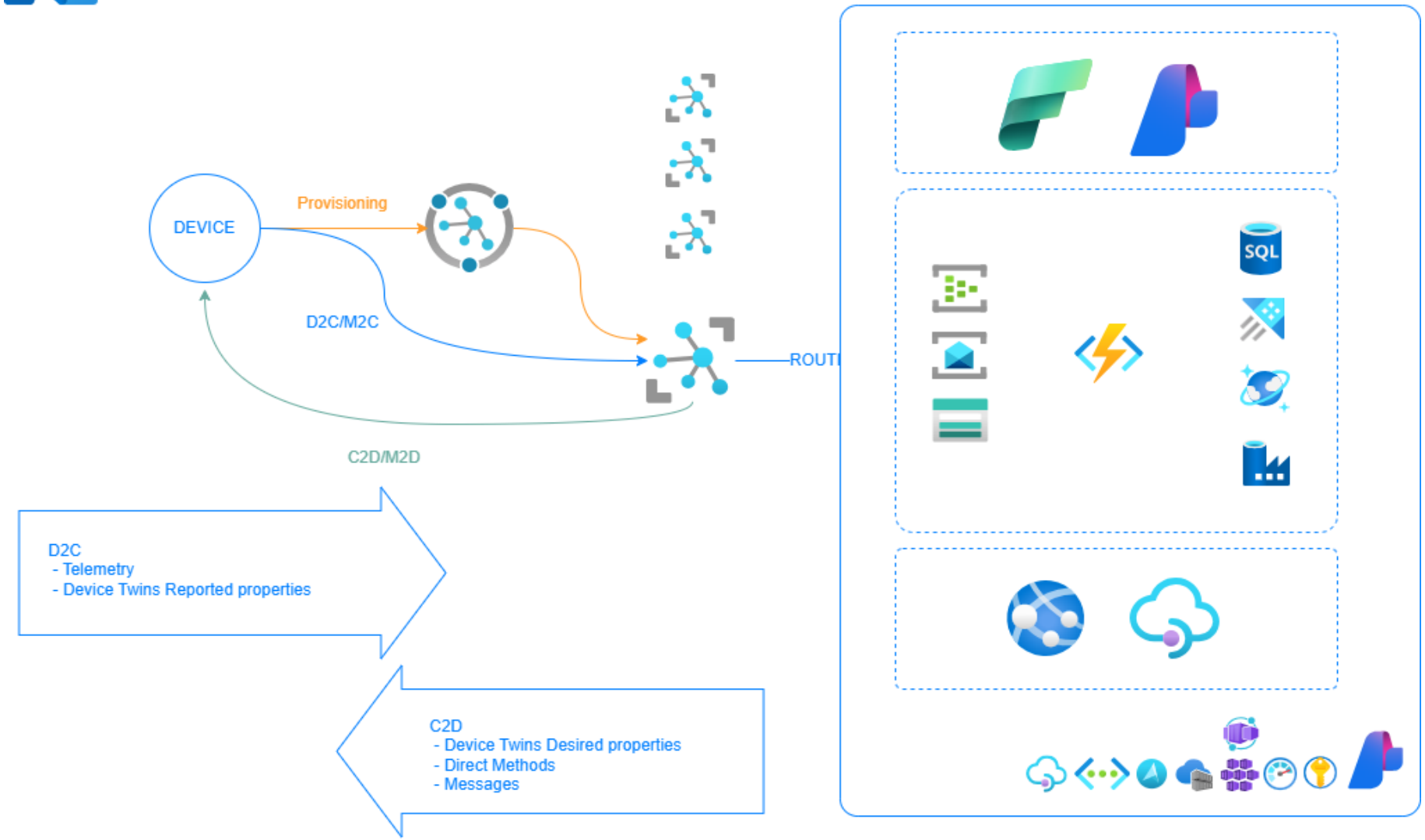
IoT Solutions – Global view



IoT Solutions – Azure IoT



IoT Solutions – Device view



IoT Flows reminder

- D2C, Device To Cloud
  - Telemetry or other (warnings, alerts, etc)
  - Twin reported properties
- C2D, Cloud To Device
  - Direct Methods
  - Messages
  - Twin Desired Properties

Azure IoT – main services (not exhaustive)

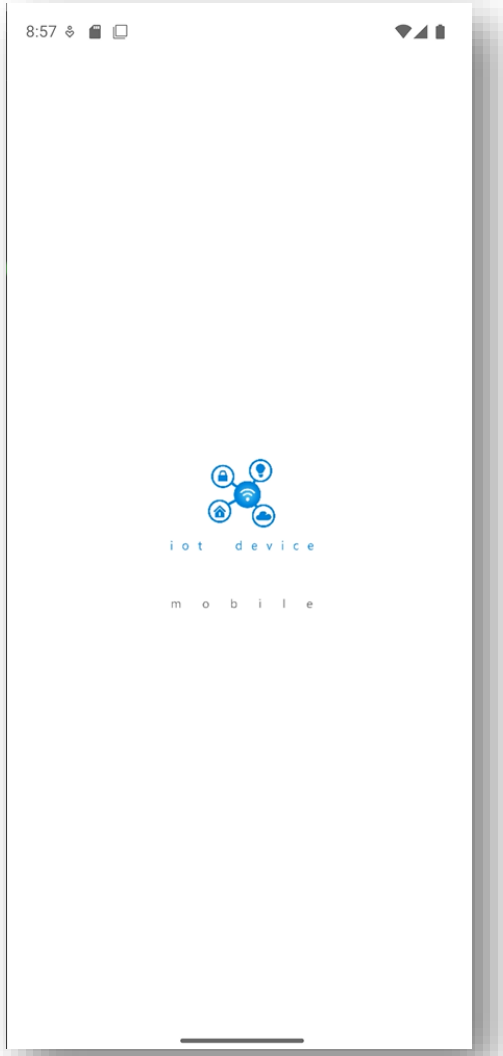
- Azure IoT Hub
- Azure IoT Hub Device Provisioning Service
- Azure Device Registry (New)
- Azure IoT Hub Update Service
- Azure IoT Operations
- Azure IoT Edge

- Azure Event Hub, Azure Service Bus, Azure Event Grid
- Azure Stream Analytics
- Microsoft Fabric (Data Factory, Analytics Databases, Real-Time Intelligence, IQ, Power BI)
- Azure Data Explorer, Azure Storage, Azure Database, Azure ComosDB
- Azure Maps

IoT Device Simulator – The “Why”?

- IoT Solutions developers’ “needs”:
  - Have a way to have different devices with different DPSs
  - Have a way to validate different types of provisioning, DPS configurations (default Twin properties) and all the flows (desired/reported properties)
  - Have a way to send different types of telemetry (schemas)
  - Have a way to validate different types of interactions through the different solutions and according to specific settings in each solution

The App



Google Play

Simulators-Android-Main Features

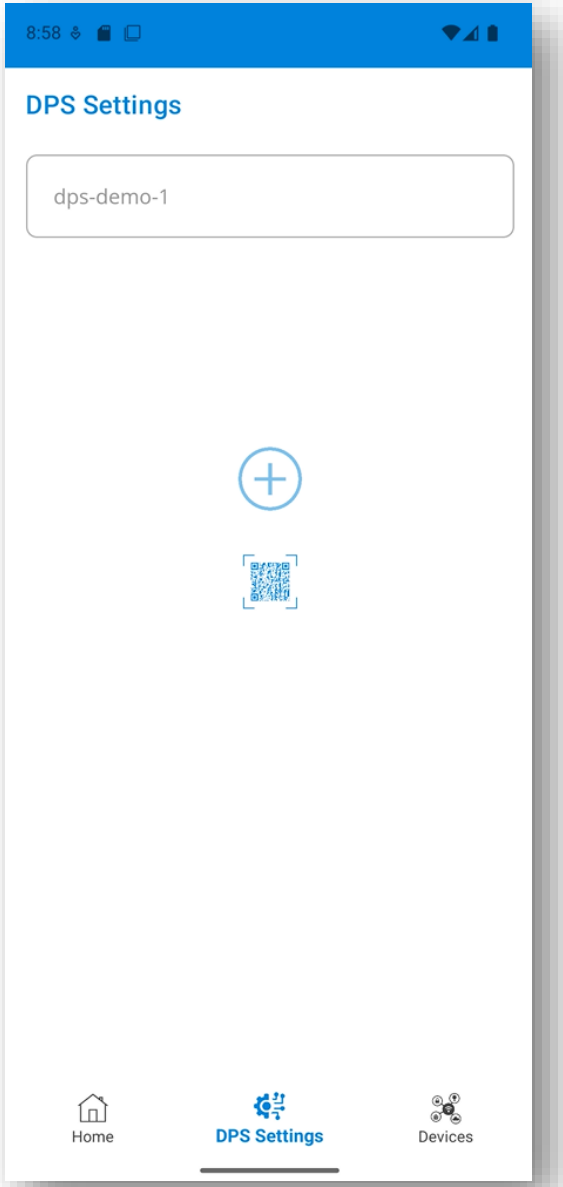
- Add as many DPSs as you want
- Provision as many devices as you want
- Each device has its own simulation settings
- Most of the D2C/C2D features in the IoT C# SDK are implemented
  - Provisioning (SAS and X509)
  - Telemetry (with random content based on a configurable JSON Schema, and configurable rate)
  - Device Twin Reported properties
  - Device Twin Desired properties
  - Direct Methods (generic and some specific like SetTelemetryInterval)

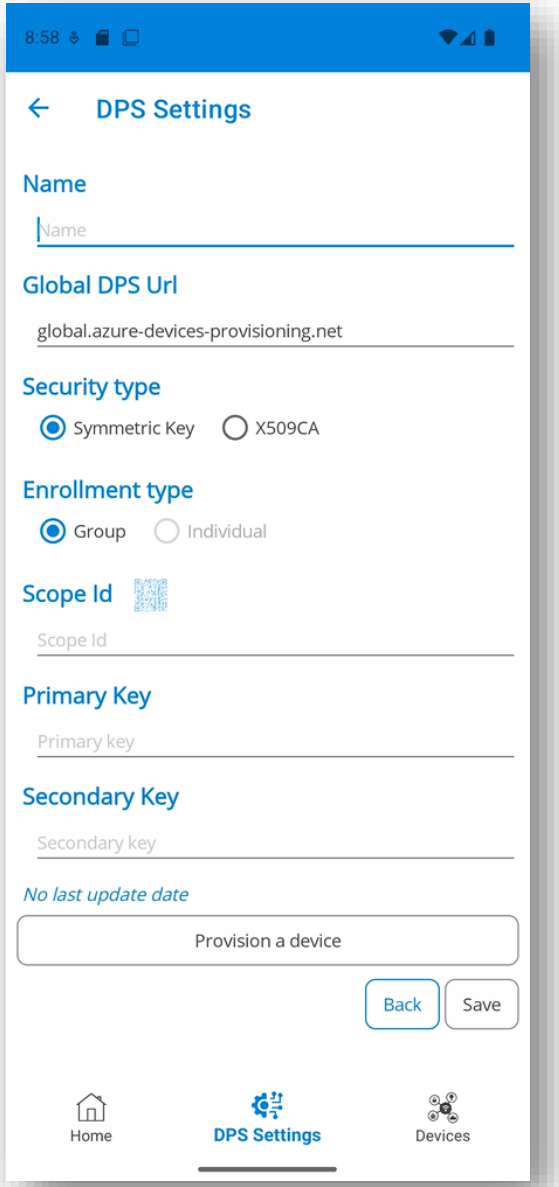
  https://github.com/jonmikeli/docs/blob/main/iot/tools/mobile-app/iot-simulator-mobileapp.md#telemetry-interval
  - C2D Messages

Technical aspects

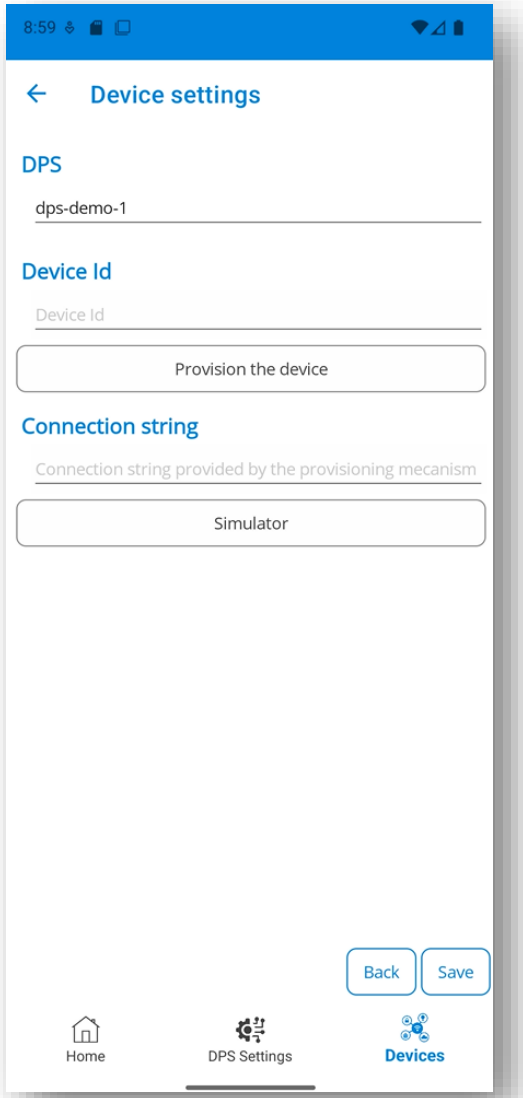
- MAUI - Android
- .NET 10
- Integrates a “light” version of a richer implementation of the simulator

Add a DPS

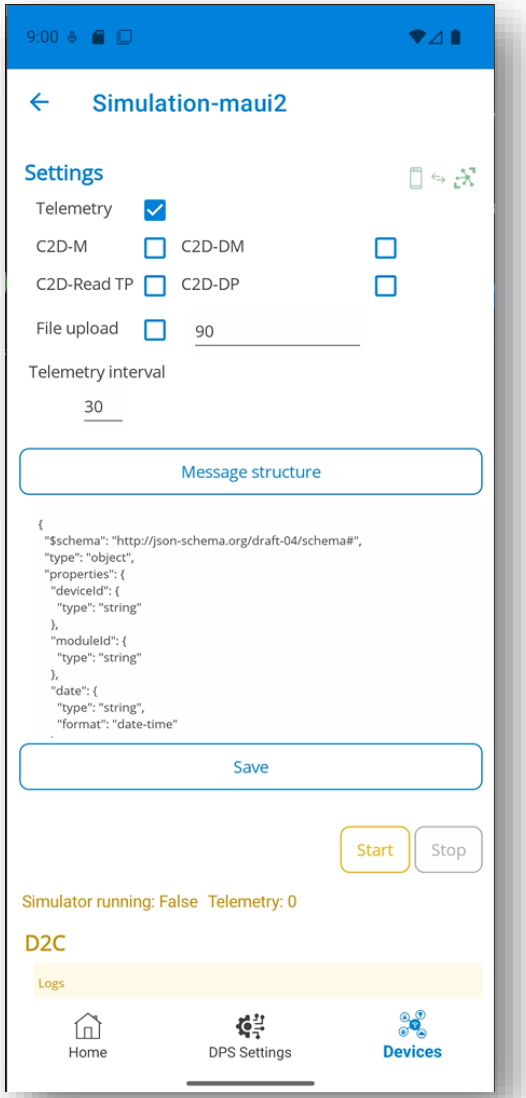

8:58
DPS Settings
dps-demo-1
Home
DPS Settings
Devices


8:58
DPS Settings
Name
Name
Global DPS Url
global.azure-devices-provisioning.net
Security type
Symmetric Key
X509CA
Enrollment type
Group
Individual
Scope Id
Scope Id
Primary Key
Primary key
Secondary Key
Secondary key
No last update date
Provision a device
Back
Save
Home
DPS Settings
Devices

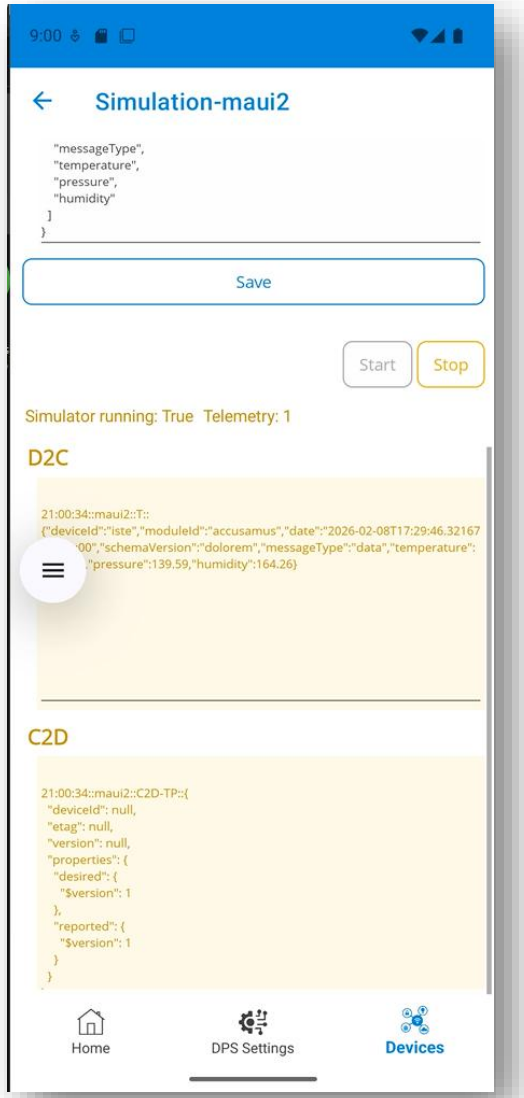
Provision a device



Configure simulation settings



Start simulation



Demo1

Mobile App

Simulators – All the different types

- Mobile
  - Regular
  - Teltonika version
- Containerized
  - Single device (Docker Hub)
  - Multidevice (Docker Hub)
  - IoT Aspire solution
- Others in progress (ex: MCP Server for the simulators)

Demo2

Container

Demo3

Aspire

Q/A

Thank you

Additional resources

Resources

- Google Play link
  - https://play.google.com/store/apps/details?id=jmi.iot.device.mobile2

- Detailed documentation
  - https://github.com/jonmikeli/docs/blob/main/iot/tools/mobile-app/iot-simulator-mobileapp.md
- GitHub
  - https://github.com/jonmikeli/mobile-iot-client-net10-shared
- Azure IoT Hub
  - https://azure.microsoft.com/fr-fr/products/iot-hub
- IoT Hub SDK C#
  - https://github.com/Azure/azure-iot-sdk-csharp

Container-Single device

- Same features as mobile app
- Plus:
  - cli approach (making it industrializable)
  - More robust X509 integration
- Container:

docker pull jonmikeli/iot-simulator-dps:10.0.892

- Documentation: https://github.com/jonmikeli/docs/blob/main/iot/tools/simulators/iot-simulator-single-device-container.md

Container-Multi device

- Same features as mobile app
- Plus:
  - cli approach (making it industrializable)
  - No X509 features
- Container:

docker pull jonmikeli/iot-simulator-dps-multi:10.2.867

- Documentation:

https://github.com/jonmikeli/docs/blob/main/iot/tools/simulators/iot-simulator-multi-device-container.md

Aspire-Multi device

- .NET CLI and Visual Studio Extension (VSIX) to create such solutions
- NuGet
  - https://www.nuget.org/packages/JMI.Aspire.IoTSimulator.DPS.SingleDevice.Template/
  - https://www.nuget.org/packages/JMI.Aspire.IoTSimulator.DPS.MultiDevice.Template/
- VSIX
  - https://marketplace.visualstudio.com/items?itemName=JonMikelInza.AspireIoTSimulatorDPSSingleDeviceTemplate
  - https://marketplace.visualstudio.com/items?itemName=JonMikelInza.AspireIoTSimulatorDPSMultiDeviceTemplate

Device Simulator Git Hub

- https://github.com/jonmikeli/mobile-iot-client-net10-shared